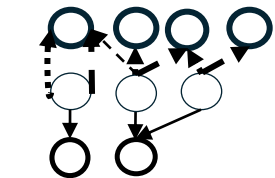
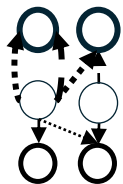




Demonstrations



P_1 .Age

...

...

...

...

P_4 .Age

...

...

...

...

...

P_5 .Age

...

...

...

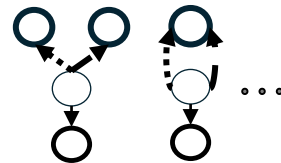
...

...



Graph Mining

Subgraph discovery under constraints



Pattern recovery from graphs

```
(n1: Dept) <- [e1:Work] - (n2: Faculty) -  
[e2:Teach] -> (n3: Course),  
(n2) - [e3:Taught] -> (n4: Course)  
...
```



Sketch Generation

```
MATCH (n1: Dept) <-  
[e1:Work] - (n2: Faculty) ...  
WHERE ?  
RETURN ?1.Age, ...
```

Sketch Completion

Constraint-based optimization

FILTER clause completion

Enumeration with deductive reasoning

Enumeration on RETURN clause and requirements for FILTER clause

```
MATCH (n1: Dept) <- [e1:Work] - ... WHERE include G1, G2, G4, G5 and exclude G3  
RETURN n2.Age, n2.Title, e1.Years, ...
```



Query